

GRAVITATION

Complete Formula Handbook

Newton's Law · Gravitational Field & Potential
Kepler's Laws · Satellites · Escape & Orbital Velocity

For JEE Main / Advanced & NEET Physics

1. Newton's Law of Gravitation

Every particle of matter attracts every other particle with a force directly proportional to the product of their masses and inversely proportional to the square of the distance between them.

$$F = G m_1 m_2 / r^2$$

$G = 6.674 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$ (Universal Gravitational Constant)

Vector Form

$$\mathbf{F} = -G m_1 m_2 / r^2 \text{ (along the line joining the masses)}$$

2. Acceleration Due to Gravity (g)

$$g = GM / R^2$$

At Earth's surface; M = mass of Earth, R = radius of Earth

Variation with Height (h)

$$g_h = g (1 - 2h/R) \text{ [for } h \ll R]$$

Exact: $g_h = g R^2 / (R+h)^2$

Variation with Depth (d)

$$g_d = g (1 - d/R)$$

g decreases linearly with depth; $g = 0$ at Earth's center

Variation due to Rotation of Earth (latitude λ)

$$g' = g - \omega^2 R \cos^2 \lambda$$

Maximum decrease at equator ($\lambda=0$), no change at poles ($\lambda=90^\circ$)

3. Gravitational Field & Intensity

Gravitational field intensity is the force experienced by a unit mass placed at a point in the gravitational field.

$$\mathbf{E} = \mathbf{F}/m = \mathbf{GM} / r^2$$

Directed towards the source mass; SI unit: N/kg

Field due to a Uniform Solid Sphere

- Outside ($r \geq R$): $E = GM/r^2$
- On surface ($r = R$): $E = GM/R^2$
- Inside ($r < R$): $E = GMr/R^3$ (increases linearly with r)

4. Gravitational Potential (V)

Gravitational potential at a point is the work done in bringing a unit mass from infinity to that point, without acceleration.

$$V = -GM / r$$

Always negative; zero at infinity; SI unit: J/kg

Potential due to a Uniform Solid Sphere

- Outside/Surface ($r \geq R$): $V = -GM/r$
- Inside ($r < R$): $V = -GM(3R^2 - r^2) / 2R^3$
- At center ($r = 0$): $V = -3GM/2R$ ($= 1.5 \times$ surface value)

Relation between E and V

$$E = -dV/dr$$

5. Gravitational Potential Energy (U)

$$U = -GMm / r$$

Work done by external agent to move mass m from r to infinity = $-U$

PE near Earth's surface (small height h)

$$U = mgh$$

Valid only when $h \ll R$ (uniform g approximation)

6. Escape Velocity

Minimum velocity needed for an object to escape a planet's gravitational field permanently, found by setting total mechanical energy to zero.

$$v_e = \sqrt{2GM/R} = R\sqrt{8\pi G\rho/3} = \sqrt{2gR}$$

For Earth: $v_e \approx 11.2$ km/s

7. Kepler's Laws of Planetary Motion

Law	Statement	Formula
1. Law of Orbits	Planets move in elliptical orbits with the Sun at one focus.	—
2. Law of Areas	The radius vector sweeps equal areas in equal time intervals (areal velocity constant).	$dA/dt = L/2m = \text{constant}$
3. Law of Periods	Square of time period is proportional to cube of semi-major axis.	$T^2 \propto a^3$

$$T^2 = (4\pi^2 / GM) \cdot a^3$$

Derived form of Kepler's Third Law

8. Orbital Velocity of a Satellite

Speed required for a satellite to maintain a stable circular orbit, where gravitational force provides the necessary centripetal force.

$$v_o = \sqrt{GM/r} = R\sqrt{g/(R+h)}$$

For orbit close to surface ($h \approx 0$): $v_o = \sqrt{gR} \approx 7.9 \text{ km/s}$

Relation between Orbital and Escape Velocity

$$v_e = \sqrt{2} \cdot v_o$$

9. Time Period of a Satellite

$$T = 2\pi r / v_o = 2\pi\sqrt{r^3/GM}$$

Geostationary Satellite

Orbits in the equatorial plane, from west to east, with the same time period as Earth's rotation ($T = 24 \text{ h}$), appearing stationary relative to a point on Earth.

$$r \approx 4.2 \times 10^4 \text{ km (height } \approx 36,000 \text{ km above surface)}$$

10. Energy of a Satellite in Orbit

Quantity	Formula
Kinetic Energy (KE)	$+ GMm / 2r$
Potential Energy (PE)	$- GMm / r$
Total Energy (TE)	$- GMm / 2r$
Binding Energy	$+ GMm / 2r (= -TE)$

Note: Total energy is negative for a bound system, indicating the satellite is gravitationally bound to the planet. As $r \rightarrow \infty$, $TE \rightarrow 0$, marking the escape condition.

11. Quick Revision Table — Key Relations

Quantity	Formula	SI Unit
Gravitational Force	$F = Gm_1m_2/r^2$	N
Field Intensity	$E = GM/r^2$	N/kg
Potential	$V = -GM/r$	J/kg
Potential Energy	$U = -GMm/r$	J
Escape Velocity	$v_e = \sqrt{2GM/R}$	m/s
Orbital Velocity	$v_o = \sqrt{GM/r}$	m/s
Time Period	$T = 2\pi\sqrt{r^3/GM}$	s